David Xia

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Research Interests

Reinforcement learning, foundation models, LLM agents, RL theory

Education

- **University of Illinois Urbana-Champaign** *Aug 2023 - May 2027*
  Bachelor of Science in Computer Science Champaign, Illinois
  Bachelor of Science in Liberal Arts & Sciences Major in Mathematics, Data Optimization Concentration
  Bachelor of Science in Liberal Arts & Sciences Major in Statistics
  - GPA: 3.99/4.00

Papers

REV=Under Review, ACC=Accepted

2026 **[REV] Benchmarking agentic optimizers**
Co-first author. Title and author list withheld during double-blind review.
Under review at ICLR 2027

2026 **[ACC] RSK linear operators and the Vershik-Kerov-Logan-Shepp curve** [Journal] [arXiv]
Duy Phan, **David Xia**
Electronic Journal of Combinatorics, 33(3), #P17

2025 **[ACC] EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations** [IEEE] [arXiv]
Jung-Woo Chang, Ke Sun, **David Xia**, Xinyu Zhang, Farinaz Koushanfar
IEEE Symposium on Security and Privacy, 2025

Workshop Papers, Manuscripts & Notes

NOTE=Technical Note, SUB=Submitted, ACC=Accepted

2026 **[NOTE] Frozen Policy Iteration for MDPs with Stochastic Transitions under Linear Q^π Realizability** [PDF]
David Xia, Ruizhong Qiu, Hanghang Tong
Technical note

2026 **[SUB] Predictability in Major League Sports: Betting Odds versus Mathematical Models** [PDF]
David Xia, AJ Hildebrand
Submitted to the Mathematics and Sports (MAS) Journal

2026 **[ACC] Revitalizing Local Democracy: A Human-Centered Audit of LLMs in City Council Journalism** [PDF]
David Xia, Chris Maury
Human-centered Evaluation and Auditing of Language Models (HEAL) Workshop at CHI 2026

Research Experience

- **Northwestern University**, Research Intern *May 2026 - present*
  - Advised by Professor Zhaoran Wang.
  - Research on LLM agents as black-box optimizers: how agents propose candidate solutions, interpret feedback, and revise their search strategy across iterations.
  - Co-first-authored paper under double-blind review at ICLR 2027. Further details withheld during the review period.

- **iDEA-iSAIL Lab**, Research Intern *Feb 2026 - present*
  - Advised by Professor Hanghang Tong.

- Conducted research in theoretical reinforcement learning, studying sample and computation efficient algorithms for online learning in finite-horizon Markov Decision Processes under function approximation and stochastic dynamics.
- Used concentrability to develop the first sample and computation efficient algorithm for online reinforcement learning in stochastic-transition MDPs under the linear Q^π realizability assumption, resolving an open problem left by prior work restricted to deterministic dynamics.
- Established a PAC sample complexity guarantee of $\tilde{O}(d^2H^7C^*/\varepsilon^3)$ by combining techniques from both offline and online reinforcement learning theory, dynamic programming, concentration analysis, and linear function approximation.
- Technical note.

• **Illinois Combinatorics Lab for Undergraduate Experience (ICLUE)**, Research Intern *Aug 2024 - May 2026*
- Advised by Professor Alexander Yong.
- Proved a new asymptotic result that the probability that the Schensted insertion algorithm for a uniformly random permutation in S_n exhibits a special "bumping" interaction converges to 1 as n approaches infinity.
- Derived the result using rigorous probability bounds on the RSK linear operator by leveraging the Vershik-Kerov-Logan-Shepp limit shape curve as the central analytic tool.
- Developed proofs requiring techniques spanning real analysis (uniform convergence, limit arguments), probability theory (concentration inequalities, convergence in probability), and asymptotic analysis (bounding combinatorial quantities in the large-n regime).
- Paper published in the Electronic Journal of Combinatorics, 33(3), #P17.

• **Illinois Mathematics Lab**, Research Intern *Jan 2024 - May 2026*
- Advised by Professor AJ Hildebrand.
- Researched the data science perspective of sports matches, analyzing the predictability of outcomes by comparing prediction accuracies across polls, the betting market, Elo ratings, and mathematical models like Bradley-Terry.
- Presented findings at Rose-Hulman Undergraduate Math Conference 2024, UIUC Undergraduate Research Symposium 2024, and Joint Mathematics Meeting 2025.
- Paper in submission to the Mathematics and Sports (MAS) Journal.

• **Carnegie Mellon University, HCII**, Research Intern *May 2025 - Feb 2026*
- Advised by Professor Jeff Bigham.
- Designed and evaluated an automated journalism pipeline using SOTA LLMs to replicate end-to-end editorial workflows, from transcript segmentation to headline generation and topic prioritization.
- Orchestrated large-scale crowd-sourced study and demonstrated that LLM-based headline quality and topic prioritization can meet and exceed professional standards.
- Paper in CHI 2026 Human-centered Evaluation and Auditing of Language Models (HEAL) workshop.

• **National University of Singapore, SERIUS REU**, Research Intern *May 2024 - Aug 2024*
- Advised by Professor Kelvin Fong Xuanyao.
- Designed the GARDEN (Generalized Anomaly Recognition and Detection for Enhanced Nurturing) pipeline for early disease detection in farm plants, combining segmentation, generative, and classification models end-to-end.
- Built a leaf segmentation model to isolate regions of interest from complex backgrounds, enabling downstream anomaly detection to focus on plant tissue only.
- Implemented a conditional GAN (cGAN) to synthesize realistic diseased and healthy leaf images, augmenting training data and improving classifier generalization.
- Achieved robust disease classification by ensembling deep learning and traditional ML models, exploiting complementary strengths to improve detection reliability on real farm images.

• **University of California San Diego**, Research Intern *Jun 2023 - Dec 2024*
- Advised by Professor Xinyu Zhang.

- Developed EveGuard, a software-based defense framework to protect voice privacy from vibrometry-based side channels, using a perturbation generator model (PGM) to suppress sensor-based eavesdropping while preserving high audio quality.
- Implemented Eve-GAN, a novel domain-translation task for inferring eavesdropped signals, enabling end-to-end training of the PGM and using few-shot learning to reduce data-collection overhead.
- Achieved a protection rate of over 97% against audio classifiers, hindering eavesdropped reconstruction.
- Paper published in the IEEE Symposium on Security and Privacy 2025.

Relevant Coursework

Highlighted courses are graduate-level.

In progress (Fall 2026): Statistical Reinforcement Learning, Randomized Algorithms
CS: Machine Learning, Deep Learning, Natural Language Processing, Algorithms, Numerical Analysis, Numerical Methods
Math: Honors Real Analysis, **Algebraic Combinatorics**, Abstract Linear Algebra, Combinatorics, Graph Theory, Abstract Algebra, Complex Variables
Stats: Bayesian Analysis

Grants and Awards

2025 CMU HCII REU: NSF 2349558

2025 ICLUE RTG: NSF 1937241

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